



# Smart Blood Bank and Donor Locator Management System

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## Abstract

- Web-based platform for managing blood donation and blood requests, helping patients and hospitals find donors by blood group and location.
- Built on MySQL / PostgreSQL and MongoDB to cut search time and improve blood availability management.

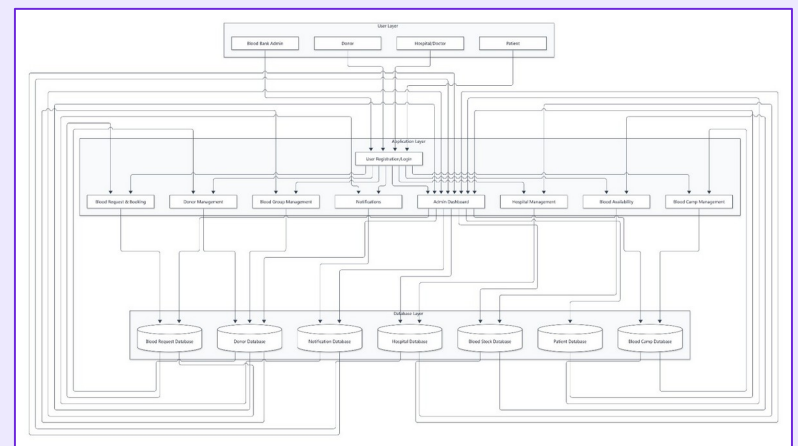
## Tools & Technologies

- Database (RDBMS):** MySQL / PostgreSQL
- NoSQL Database:** MongoDB
- Language:** SQL (queries, joins, transactions)
- Tools:** VS Code, MySQL Workbench, MongoDB Compass
- Version Control:** Git & GitHub
- Front-End:** Flutter (web app)

## Problem Statement & Objectives

- Problem:** Finding the required blood group and nearby compatible donors in an emergency is slow because data is scattered and manual.
- Objective 1:** Centralized donor and blood-bank management
- Objective 2:** Quick donor search by blood group and location
- Objective 3:** Online blood requests with emergency support
- Scope:** Donors, patients, hospitals, blood banks, requests and stock; RDBMS + MongoDB

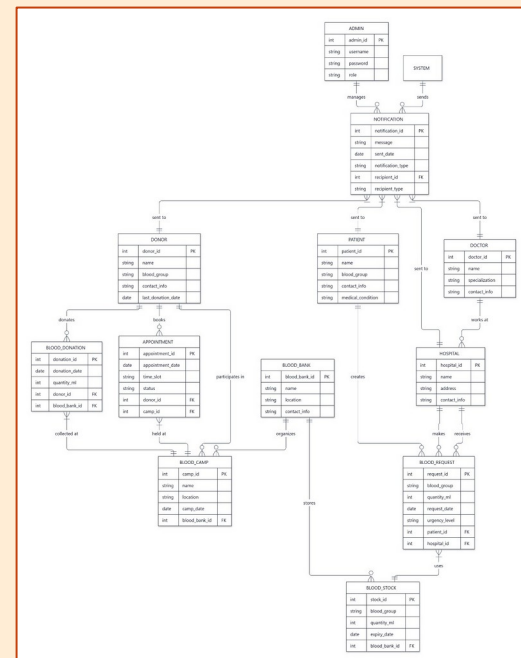
## System Architecture



## Existing vs Proposed System

| Feature            | Existing System                           | Proposed System                                  |
|--------------------|-------------------------------------------|--------------------------------------------------|
| Blood Availability | Manual checks with blood banks            | Centralized online availability check            |
| Donor Search       | Registers / separate databases            | Blood-group + location based search              |
| Emergency Requests | Phone calls, personal contacts            | Online requests with emergency support           |
| Records            | Scattered, hard to keep updated           | Centralized donor and stock records              |
| Limitation / Gap   | No central view; delays in finding donors | Blood-group matching + location search cuts time |

## ER Diagram & Database Design



### 13 Entities

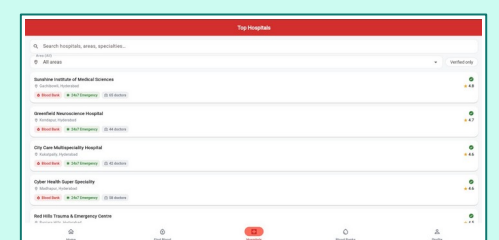
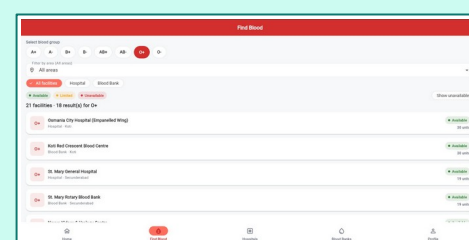
Donor  
Patient  
Doctor  
Hospital  
Blood Bank  
Blood Camp  
Blood Stock  
Blood Request  
Blood Donation  
Appointment  
Notification  
Admin  
System

PK – primary key  
FK – foreign key

## Feasibility Study

- Technical:** Open-source RDBMS and MongoDB, SQL skills; VS Code, Workbench, Compass and Git available.
- Economic:** Free, open-source stack, so no licensing cost.
- Operational:** Simple search and request flow; easy to use and maintain.
- Schedule:** RDBMS and NoSQL modules fit within the semester timeline.

## Implementation Screenshots



## Results & Testing

- ER model: 13 entities, PKs and FKs
- 3-layer architecture designed
- Working UI: blood search, hospitals
- Planned: CRUD, joins, transactions

## Conclusion

- Centralized DBMS platform for donors, requests and blood-bank inventory
- Finds donors by blood group and location, cutting emergency delays

## Future Scope

- GPS-based real-time donor locator
- Emergency request notifications
- Mobile app and cloud deployment
- Analytics and demand prediction